This document reviews the most important concepts needed for Exam 1. This is just a handy reference while studying the practice problems. You will need to know the formulas we use for the exam itself.

1 First-Order Equations

Section 1.2: Solving equations with integrals

If a differential equation is of the form

$$\frac{dy}{dx} = f(x)$$

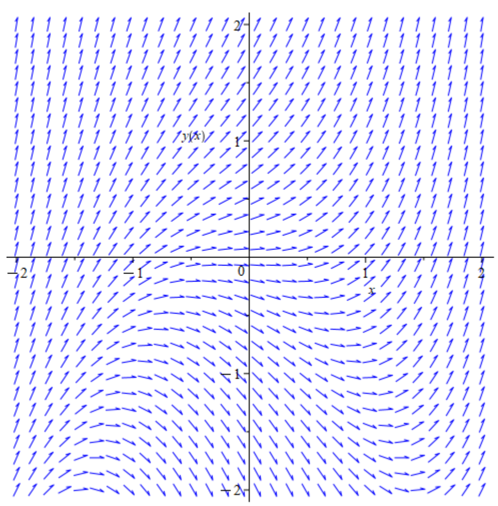
then we can solve for $y(x)$ simply by integrating:

$$y(x) = \int f(x)\,dx$$

The integral will contain an arbitrary constant $+C$.

Section 1.3: Slope Fields

Given a differential equation $\frac{dy}{dx} = f(x, y)$ we can create a plot of small slope segments where at each point (x, y) in the plane, we draw a small segment (or arrow) with slope $f(x, y)$. Here is an example for $\frac{dy}{dx} = x^2 + y$:



A good strategy to identify slope fields is to take the equation $\frac{dy}{dx} = f(x, y)$ and figure out where $f(x, y) > 0$, $f(x, y) = 0$, and $f(x, y) < 0$. In our example, the slope is zero when

$y = -x^2$, and we can see this parabola in the slope field with (at least close to) horizontal segments forming the curve. The slope is positive when $y > -x^2$ and that's everywhere above the parabola $y = -x^2$. The slope is negative when $y < -x^2$, below the parabola. Simply put, finding where the slope is positive, negative, and zero can be very telling for a slope field.

Section 1.4: Separable equations

If a differential equation is **separable** it is of the form

$$\frac{dy}{dx} = g(x)h(y),$$

that is, we can separate the x and y expressions, then we can write the formal integrals

$$\int \frac{dy}{h(y)} = \int g(x)\,dx.$$

If we can compute both integrals, we will end up with an implicit solution of the form $F(x, y) = C$ for some arbitrary constant C. Whether or not we can solve explicitly for y depends on the expression $F(x, y)$.

Section 1.5: First-order linear equations

A first-order linear equation can be written in the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

for some functions P and Q. We can define a new function ρ by the rule

$$\rho(x) = e^{\int P(x)\,dx}.$$

In this integral, it is okay to ignore $\pm$ that may appear in front of the function, and the $+C$ from the P integral. With this ρ (called an integrating factor), we can then solve the differential equation with the formula

$$y = \frac{\displaystyle\int \rho(x)Q(x)\,dx}{\rho(x)}.$$

Note: the integral in this expression requires a $+C$ in order to account for the initial condition placed on y.

Section 1.5 Application

For a mixing problem, if we are trying to solve for the amount (mass) of a solvent $x(t)$, like salt, at time t in a tank filled with a solution of volume $V(t)$, the tank is being filled with fluid of concentration c_i and flow rate r_i, while the well-mixed solution is leaving the tank at a rate r_o and concentration c_o, then we can model the problem with the differential equation

$$\frac{dx}{dt} = r_i c_i - r_o c_o.$$

The well-mixed assumption implies $c_o = \dfrac{x(t)}{V(t)}$ and, assuming the rates r_i and r_0 are constant, we have $V(t) = (r_i - r_o)t + V_0$. Thus, the initial value problem to solve is

$$\frac{dx}{dt} = r_i c_i - r_o \frac{x}{(r_i - r_o)t + V_0}, \qquad x(0) = x_0.$$

Section 1.6: Substitutions

Given a differential equation $\dfrac{dy}{dx} = f(x, y)$, we may try to make a substitution with a new variable v to simplify the expression. For example,

$$\frac{dy}{dx} = (y - x + 2)^3 + 1$$

we could try to simplify by letting $v = y - x + 2$ and then $\dfrac{dv}{dx} = \dfrac{dy}{dx} - 1$ which gives

$$\frac{dv}{dx} = v^3.$$

The equation is now solved as separable equation in terms of v and x. We can use $v = y-x+2$ to go back to y and x. In general, the substitution will be straightforward to spot, like in this example, or you will be told what the substitution for v is!

Some instances have a known, useful substitution provided we know some of the structure. Bernoulli equations take the particular form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, \qquad n \neq 0, 1.$$

These equations are nonlinear, but all the nonlinear essence is consolidated into the power y^n. If we have this equation, the substitution $v = y^{1-n}$ will transform this equation into a first-order linear equation to solve for v. Once we find v, the original substitution tells us how to go back to y. As an example, consider

$$xy^2 \frac{dy}{dx} + x^2 y = y^3$$

which can be rewritten as

$$\frac{dy}{dx} + xy^{-1} = \frac{y}{x}.$$

This equation is Bernoulli with $n = -1$. So, $v = y^{1-(-1)} = y^2$. Differentiation gives $\frac{dv}{dx} = 2y\frac{dy}{dx}$. Using the original differential equation gives

$$\frac{dv}{dx} = 2y\frac{dy}{dx} = 2y\left(-xy^{-1} + \frac{y}{x}\right) = -2x + \frac{2y^2}{x}$$

$$\iff \frac{dv}{dx} - \frac{2}{x}v = -2x.$$

This equation is now first-order linear in terms of v and x.

Section 1.6: Exact Equations

If we are given a diffferential equation of the form

$$M(x,y) + N(x,y)\frac{dy}{dx} = 0$$

or equivalently

$$M(x,y)dx + N(x,y)dy = 0$$

the equation is said to be **exact** if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. If the equation is exact, then we can find a function F that satisfies the **partial integrals**

$$F(x,y) = \int M(x,y)\,dx$$

$$F(x,y) = \int N(x,y)\,dy$$

Once we find $F(x,y)$, the general (implicit) solution to the exact equation is $F(x,y) = C$.

2 Applications and Approximations

Section 2.1: Population Models

We should be familiar with the general logistic equation

$$\frac{dP}{dt} = kP(M-P)$$

and how to solve using partial fractions. The general solution is $P(t) = \dfrac{P_0 M}{P_0 + (M - P_0)e^{-kMt}}$ where $P(0) = P_0$. This is rather complicated, and it is better to just solve the equation from scratch when the need arises. Especially considering that there are similar looking equations like

$$\frac{dx}{dt} = x^2 - x$$

where the square term has a $+$ instead of a $-$ which will change the solution (and behavior) slightly. Typically, since these equations are autonomous, we can usually infer a lot about them from the techniques in section 2.2.

Section 2.2: Autonomous Equations and Stability

Given an autonomous equation $\dfrac{dy}{dx} = f(y)$ (the right hand side only depends on the dependent variable y) we can find the equilibrium solutions (critical points) by finding where $f(y) = 0$ (i.e. where the derivative is 0). If $f(a) = 0$, then the constant valued solution $y(x) = a$ satisfies the differential equation (hence why it's called an equilibrium solution). We can then construct a diagram and note whether the derivative is positive (indicated with a rightward arrow) or negative (indicated with a leftward arrow) away from the equilibrium solutions. Review the lecture notes on this section for description of the three types of stability: Stable, Unstable, Semistable.

Section 2.4: Euler's Method

Suppose we are given an initial value problem $\dfrac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$ and we want to approximate a value $y(x_n)$ such that, for some fixed step size $h > 0$, $x_n = x_0 + nh$. We can generate approximations with the algorithm

$$\begin{gathered}
x_1 = x_0 + h, \quad y_1 = y_0 + hf(x_0, y_0) \\
x_2 = x_1 + h, \quad y_2 = y_1 + hf(x_1, y_1) \\
\vdots \\
x_n = x_{n-1} + h, \quad y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})
\end{gathered}$$

After we finish these computations, then $y(x_n) \approx y_n$. The specific formulas come from a repeated application of using the tangent line to approximate nearby values (See the notes on Section 2.4).

3 Higher Order Differential Equations

In general, given an n-th order linear homogeneous equation with constant coefficients

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 = 0$$

we can write the characteristic equation

$$a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = 0$$

which will have n roots $r_1, r_2, \ldots, r_n$ counting multiplicity and allowing complex number. We isolate each distinct root r and proceed as follows:

r is a real root with multiplicity m

We choose the m functions

$$e^{rx}, xe^{rx}, x^2 e^{rx}, \ldots, x^{m-1} e^{rx}$$

Note: This still works when $m = 1$ (i.e. the root occurs exactly once). In this case, the above expression just gives e^{rx}

r is a complex valued root $r = a + ib$ with multiplicity m

Then, the complex conjugate $\overline{r} = a - ib$ must also be a root among $r_1, \ldots, r_n$ and it must also have multiplicity m to match r (complex roots come in complex conjugate pairs since our coefficients to the differential equation are real numbers). Although we could use the same approach above to get m solutions for r and m solutions for $\overline{r}$ (giving $2m$ solutions in total), it is better in practice to use the following $2m$ real valued functions instead:

$$\begin{cases} e^{ax}\cos(bx), xe^{ax}\cos(bx), \ldots, x^{m-1}e^{ax}\cos(bx) \\ e^{ax}\sin(bx), xe^{ax}\sin(bx), \ldots, x^{m-1}e^{ax}\sin(bx) \end{cases}$$

After doing this procedure for all distinct values r, we can put all the functions together to write the general solution. As an example, suppose we have a *seventh* order equation

$$a_7 y^{(7)} + a_6 y^{(6)} + a_5 y^{(5)} + a_4 y^{(4)} + a_3 y''' + a_2 y'' + a_1 y' + a_0 y = 0$$

has the characteristic equation

$$a_7 r^7 + 6a_6 r^6 + a_5 r^5 + a_4 r^4 + a_3 r^3 + a_2 r^2 + a_1 r + a_0 = 0$$

with roots $r = 2+3i, 2+3i, 2-3i, 2-3i, 1, 1, -1$. The general solution can be written as

$$y = c_1 e^{2x}\cos(3x) + c_2 e^{2x}\sin(3x) + c_3 xe^{2x}\cos(3x) + c_4 xe^{2x}\sin(3x) + c_5 e^x + c_6 xe^x + c_7 e^{-x}$$

In general, most of the time we will only deal with characteristic equations of order 2:

$$ar^2 + br + c = 0 \iff r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

but we can encounter higher order. Really high orders, like this example, would require some extra information (either we are given the roots, or we are helped with the factoring). As seen in class, we can usually deal with third and fourth order equations by guessing small integer roots $r = \pm 1, \pm 2, \ldots$ and once we find an integer m such that $r = m$ is a root, we can divide $r - m$ as a factor from the characteristic polynomial. This can be repeated until we have an equation of order 2 where we can apply the quadratic equation. Also, you can just try factoring from the start.

Most of the content in sections 3.1 and 3.2 was preparation for these general results, and to give some background on the connections with linear algebra (solving systems of linear equations). The main takeaway is that these building blocks that we choose above are linearly independent. The tool we used to check linear independence is the Wronskian:

$$W(y_1, y_2) = \det \begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} = y_1 y_2' - y_2 y_1'.$$

If y_1 and y_2 solve the same second order linear homogeneous differential equation

$$y'' + p(x)y' + q(x)y = 0$$

then $W(y_1, y_2) \neq 0$ everywhere if and only if y_1 and y_2 are linearly independent. For general functions without the differential equation to back it up, all we can say is that $W(y_1, y_2) \neq 0$ at some point implies y_1 and y_2 are linearly independent (meaning in general, it's possible for $W(y_1, y_2) = 0$ everywhere, but y_1, y_2 are still independent). This idea can be generalized to higher order equations with a general Wronskian:

$$W(y_1, \ldots, y_n) = \det \begin{bmatrix} y_1 & \cdots & y_n \\ y_1' & \cdots & y_n' \\ y_1'' & \cdots & y_n'' \\ \vdots & \vdots & \vdots \\ y_1^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix}.$$

For example, to test linear independence for three functions y_1, y_2, y_3 we would hae

$$W(y_1, y_2, y_3) = \det \begin{bmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{bmatrix}.$$